Curriculum vitae

PERSONAL INFORMATION

Bharathkumar RAJAGOPAL

Tiruppur, Tamil Nadu - 641604, Republic of India

+91 8072253737

i.bharath@outlook.com

bharathkumar-rajagopal.github.io/

www.linkedin.com/in/bharath–r

Gender Male | Date of birth 28 November 2000 | Nationality Indian

JOB LOOKING FOR PhD Candidate

WORK EXPERIENCE

Jul 2024 – Present

Independent Researcher (Under Scholarly Mentorship)

- Engaged in independent research in environmental toxicology under the scholarly mentorship of Dr. Boobal Rangaswamy, Department of Biotechnology, PSG College of Arts & Science.
- Contributing substantively to the development and authorship of scientific manuscripts intended for peer-reviewed publication.

Jan 2024 – May 2024

Dissertation Trainee

CSIR - Indian Institute of Toxicology Research, Lucknow, India

- Conducted *in vitro* toxicity assessment of polystyrene nanoplastics on mammalian lung fibroblast cells
- Characterised nanoplastics using DLS and NTA; performed cytotoxicity assays (MTT, Comet assay)
- Analysed data via GraphPad; identified ROS-mediated toxicity pathways

Jul 2023 – Dec 2023

Postgraduate Research Assistant

Amity University, Noida, Uttar Pradesh, India

- Developed pesticide-degrading bacterial consortia microbeads for enhanced storage and transport
- Optimised bacterial release efficiency for environmental remediation applications

May 2023 – Jun 2023

Summer Research Intern

Centre for Bioscience & Nanoscience Research, Coimbatore, Tamil Nadu, India

- Biosynthesised silver nanoparticles using *Priestia megaterium* endophytes
- Evaluated antibacterial/anticancer properties via UV-Vis, FTIR, and SEM

EDUCATION AND TRAINING

2022–2024

MSc Biotechnology

ISCED 7

Amity University, Noida, Uttar Pradesh, India

- Advanced Microbiology
- Advanced Molecular Biology
- Comprehensive Bioanalytical Techniques
- Molecular Cell Biology
- Advanced Biochemistry
- Applied Biostatistics for Biotechnologists
- Python
- Advanced Cancer Biology
- Advanced Genetics

2019–2022 **BSc Biotechnology** ISCED 6

PSG College of Arts & Science, Coimbatore, Tamil Nadu, India

- Environmental Biotechnology
- Plant Biotechnology
- Chemistry of Biomolecules
- Molecular Biology
- Recombinant DNA Technology
- Immunology

PERSONAL SKILLS

Mother tongue: Tamil

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C1	C1	B2	B2	B2
	IELTS Band 7 (CEFR C1)				

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
Common European Framework of Reference for Languages

Technical skills

- Molecular Biology: PCR, SDS-PAGE, Western Blotting, DNA Sequencing
- Analytical Techniques: HPLC, GC-MS/MS, DLS, NTA, Spectrophotometry
- Bioinformatics: BLAST, SWISS-MODEL, Mega X, PyMol
- Programming: Python (Biopython), GraphPad Prism

Certifications

- Introduction to Professional Scientific Communication (NPTEL)
- Bionanotechnology (Association of Indian Biologists)
- Hands-on Training: HPLC & GC-MS/MS (ICFRE-IFGTB)

PUBLICATIONS

Manuscripts in Preparation

- **Rajagopal, B.**, V, Abiraj., & R, Ragunathan. (2025). Biogenic synthesis of silver nanoparticles using endophytic bacteria isolated from *Andrographis paniculata* and its potential applications as anticancer and antibacterial agents. *Manuscript in preparation.*

Manuscripts Submitted

- Sakthivel, I., Shanmugam, L., **Rajagopal, B.**, & Rangaswamy, B. (2025). Integrating kinetic models, gene circuits, and biofilm dynamics for enhanced exopolysaccharide production in nitrifying bacterial consortia. *Preprint in SSRN.* https://dx.doi.org/10.2139/ssrn.5235712 - *Manuscript Submitted*
- Kongaserry, A., Shanmugam, L., **Rajagopal, B.**, & Rangaswamy, B. (2025). Development and characterization of *Artocarpus heterophyllus* Lam. nanosuspension: A strategy to enhance bioavailability and antioxidant potential of plant bioactives *[Manuscript submitted for publication]*. *Nanotechnology for Environmental Engineering*.

Unpublished Thesis

- **Rajagopal, B.** (2024). *In Vitro Toxicity Assessment of Polystyrene Nanoplastics in Chinese Hamster Lung fibroblast cells (V-79)*. Amity University Uttar Pradesh. *Unpublished Master's thesis*

REFEREES

– **Dr. Alok Kumar Pandey**
  Senior Principal Scientist - CSIR - Indian Institute of Toxicology Research, Lucknow
  alokpandey@iitr.res.in
  Relationship – Dissertation Project Supervisor
– **Dr. Ravi Kant Singh**
  Professor - Amity Institute of Biotechnology, Amity University Uttar Pradesh.
  rksingh1@amity.edu
  Relationship – Postgraduate advisor
– **Dr. Boobal Rangaswamy**
  Assistant Professor – Department of Biotechnology, PSG College of Arts & Science, Tamil Nadu.
  boobal@psgcas.ac.in
  Relationship – Undergraduate advisor